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Abstract

Deepfake technology has rapidly evolved due to advancements in generative AI models such as GANs, diffusion models, and neural voice synthesis systems. While these technologies enable realistic synthetic media generation, they also introduce significant cybersecurity threats including identity fraud, misinformation, and biometric spoofing. Existing detection approaches often rely on single-modality analysis (visual or audio), which limits robustness against modern high-quality deepfakes. This research investigates a multimodal deepfake detection framework that integrates visual, audio, and temporal features to improve detection accuracy and generalization. The proposed system leverages convolutional neural networks, transformer-based encoders, and attention-based fusion mechanisms to identify inconsistencies across modalities. Benchmark datasets such as FaceForensics++, DFDC, Celeb-DF, and ASVspoof are considered for evaluation. The expected outcome is a more robust and generalizable detection system suitable for real-world cybersecurity applications.

Keywords: Deepfake detection, multimodal AI, cybersecurity, GAN forensics, audio-visual analysis, temporal modeling

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Özet

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List of Abbreviations

2-D **2** Dimension

3-D **3** Dimension

PF Particle **F**iltering

Chapter 1

Introduction

Stochastic approach to tracking is driven by a state-space model of the tracking object. The state of the object is estimated with a probability density function, which was for convenience assumed to be Gaussian in earlier works. Kalman filter [1], the most popular tracking algorithm with Gaussian probability density assumption, is a provably optimal estimator for linear systems. However, Kalman filter fails in cluttered scenes, occlusions, and sudden changes of state, for which the probability density of state is multimodal.

This thesis is organized as follows. In Chapter 2 we will introduce the original condensation algorithm and its extension with PFFL. We will detail our approach in Chapter 3, including the observation model, transition model and the proposal function. We will present our experimental results in Section 4 and conclude our work in Section 5.

Chapter 2

Literature Survey

2.1 Particle Filtering

We aim to track regions chosen on an object in a sequence of image frames. The states of these regions will be represented by *partitions* where the current state of partition i is denoted by x_i . In our implementation each partition has a reference image (template) and a state that consists of the partition's coordinates in the 2D image frame.

When we model state transition as a Markov chain with independent observations, the posterior probability of state for a particle is defined as:

$$p(x_i^t | \mathbf{Z}^t) \propto p(\mathbf{z}^t | x_i^t) p(x_i^t | \mathbf{Z}^{t-1}) \quad (2.1)$$

$$p(x_i^t | \mathbf{Z}^{t-1}) = \sum_{x_i^{t-1}} p(x_i^t | x_i^{t-1}) p(x_i^{t-1} | \mathbf{Z}^{t-1}) \quad (2.2)$$

Probability of the assumed state (2.1) x_i^t is proportional to the probability of reaching this state after observations $\mathbf{Z}^{t-1} = \{\mathbf{z}^0, \dots, \mathbf{z}^{t-1}\}$ and the probability of the current observation in this state. Probability of transition from the previous probability density to the assumed state (2.2) is found over the dynamic transition model and the posterior probabilities of this partition's particles from

the previous time step. Reader is referred to [2] for a complete derivation of posterior probability of state.

2.1.1 Condensation algorithm

Condensation algorithm [2] extends posterior probability density approximation to a sequence of frames. The iteration capitalizes on Markov chain assumption, i.e. the likelihood of a state depends on the probability density of the previous state, the probability of transition into the current state using the dynamic model and the current observation.

Given the posterior probability density for the previous time step $p(x_i^{t-1}|\mathbf{Z}^{t-1})$, which is represented with a set of particles and their probabilities $\{s_{ik}^{t-1}, \pi_{ik}^{t-1}\}$, $k = \{1, \dots, M\}$, condensation algorithm proceeds as follows:

1. Sample M particles from $\{s_{ik}^{t-1}, \pi_{ik}^{t-1}\}$.
2. Predict new states for particles using the dynamic model $s_{ik}^t = p(x_i^t | x_i^{t-1} = s_{ik}^{t-1})$.
3. Evaluate particles with respect to the current observation $\pi_{ik}^t \propto p(\mathbf{z}^t | x_i^t = s_{ik}^t)$.

The dynamic model in this algorithm consists of drift and diffusion. Drift is deterministic motion defined with velocity and acceleration, and is common for all particles of a partition. Diffusion on the other hand is Brownian motion; random movements that split particles at the identical state and enable search in the image space through random walk.

Chapter 3

Approach

The PFFL algorithm leverages the dependent behavior of multiple partitions on a single object. In the general sense, the interdependency of partitions can be incorporated either as constraints in the dynamic model $p(x_i^t|x_i^{t-1})$, or in choosing N particles from the posterior of N partitions through proposal function $g(\mathbf{x}^t)$. In this study we first propose a transition model that provides an estimation for dependent drift of partitions. Second, we relieve the factorization constraint in the proposal function $g(\mathbf{x}^t)$ and propose a simple and computationally efficient model for modeling the joint probabilities of partitions.

Chapter 4

Results and Discussion

Figure (4.1) presents a 2D plot of the magnitude of the proposal function for the partition at the mouth of a gourami. When the scaling factor σ_p is low, relative motion between the partitions are more restricted and the object is considered to be more rigid. The partitions move more freely against each other as the scaling factor increases.

In the first two tracking scenarios we test the capabilities of these three algorithms (the proposed algorithm, PFFL and SPOT) to deal with occlusion and clutter. In both scenarios the task is to track an object in a crowded scene of similar objects. As shown in Table (4.1), *platty* is a publicly available high-definition video of a crowded school of fish. *Skydiving* is a challenging video as it is low resolution and the skydivers' jumpsuits are very similar. In the third scenario we assess the tracking algorithms on sudden changes in transition dependencies of partitions. In this scenario we use a face video from the MMI facial expression database [3]. This video is carefully selected to exhibit relative motion and speed reversals (e.g. blinking, neutral-smile-neutral) together with rigid body motion (e.g. head movements) of the features.

Figure (4.2) compares tracking performances of the proposed algorithm, PFFL and SPOT. In this figure, average relative error over 12 tracking points is plotted against the frame numbers. Tracking error in PFFL parts from the other two algorithms starting from frame 20. At the end of the experiment, average relative

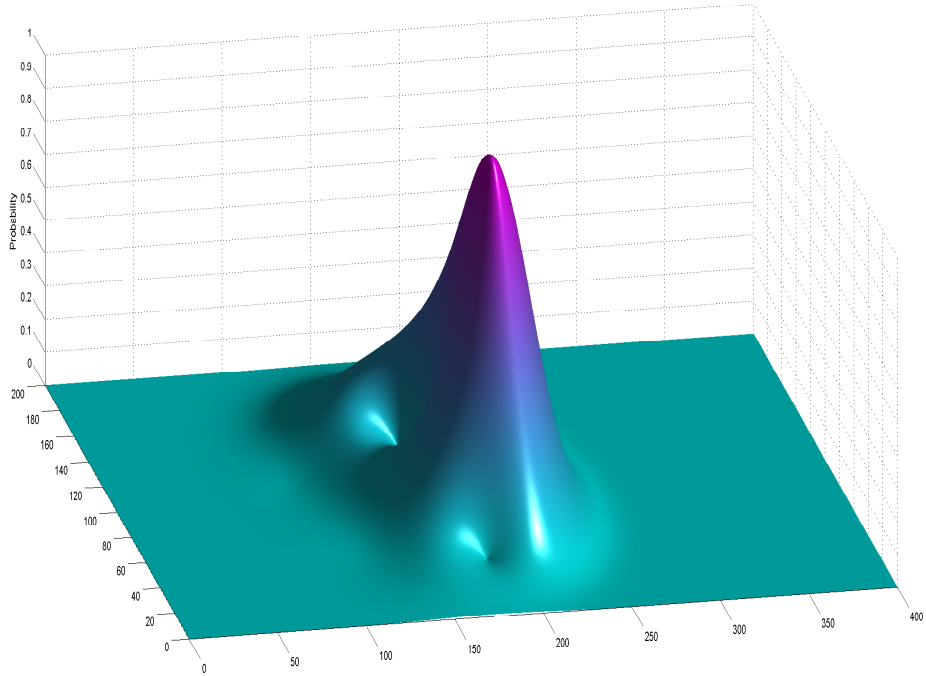


Figure 4.1: Probability field for the partition at the mouth of a gourami. Top: The magnitude of the proposal function is depicted with shades of gray. For demonstration purposes, the scaling factor is chosen higher than the actual implementation. Bottom: 3D plot of the probability field.

Video	Resolution	Frames
Platty [4]	1080×1920	110
Skydiving [5]	480×656	300
Face [3]	576×720	127

Table 4.1: Evaluation of tracking algorithms on three scenarios; platty, skydiving and face. The length of segment that is used in experiments is indicated by the number of frames.

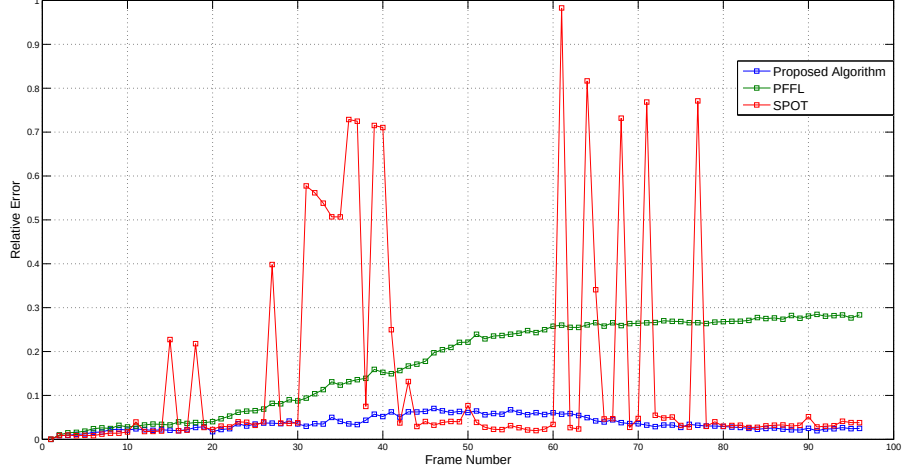


Figure 4.2: Comparison of average error on platty for the proposed algorithm (10 particles), PFFL (50 particles) and SPOT. Data markers depict the average relative error for 12 partitions for each frame.

error with PFFL is about an order of magnitude higher than the other two algorithms. SPOT occasionally drifts due to the imprecision of its observation model. Both proposed algorithm and SPOT highly benefit from the configuration based proposal function and manage to keep the general structure of platty until the end of the experiment.

Chapter 5

Conclusion

In this paper we present two contributions to tracking multiple partitions of a deformable object with particle filtering. Our first contribution is in *drift*; the deterministic motion of a partition with respect to other partitions. The second contribution is in selection of particles from probability densities of partitions such that the set of particles represent a viable and likely current state of the object.

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